Reflexivity, Symmetry, Transitivity

Properties of Congruence Modulo 3

Define relation T on $\mathbb{Z}$ as follows: $\forall m, n \in \mathbb{Z}, m\ T\ n \Leftrightarrow 3 \mid (m-n)$

1. **Reflexive?** Yes. Proof: Let $m \in \mathbb{Z}$. We must show m T m. Now *m - m = 0*. But *3 — 0* since $0 = 3 \times 0$. Hence *3 — (m - m)*. Thus by definition of T, $m\ T\ m$.

2. **Symmetric?** Yes. Proof: Left $m, n \in \mathbb{Z}, mTm$. We must show that $n\ T\ m$. By definition of T. Since $m\ T\ n$, then *3 — (m - n)*. Dy definition of divides, $\exists k \in \mathbb{Z}, m - n = 3k$. Multiplication on both sides by *-1* gives $n - m = 3(-k)$. Since $-k \in \mathbb{Z}, 3|(n-m)$. Hence, by definition of $T, n\ T\ m$.

3. **Transitive?** Hell to the yes. Proof: Let $m, n, p \in \mathbb{Z}, m\ T\ n \bigwedge n\ T\ p$. We must show $m\ T\ p$. By definition of T, since $m\ T\ n \bigwedge n\ T\ p$, then $3|(m-n) \bigwedge 3|(n-p)$. By definition of divides, $\exists r \in \mathbb{Z}, m - n = 3r \bigwedge \exists s \in \mathbb{Z}, n - p = 3s$ Adding the two equations gives $(m-n) + (n-p) = 3r + 3s$ and simplifying gives $m - p = 3(r+s)$. Since $r + s \in \mathbb{Z}, 3|(m-p)$. Hence, by definition of T, $m\ T\ p$.

Transitive Closure

Let A be a set and R a relation on A. The Transitive Closure of R is the relation R^t on A that satisfies the following three properties:

1. R^t is transitive.

2. $R \subset R^t$.

3. If S is any other transitive relation that contains R, then $R^t \subset S$

Example

Let $A = \{0, 1, 2, 3\}$ and Relation R on A is $R = \{(0,1), (1,2), (2,3)\}$. Find the transitive closure of R.
Answer: $\{(0,1), (1,2), (2,3)\} \subset R^t . R^t = \{(0,1), (0,2), (0,3), (1,2), (1,3), (2,3)\}$

Equivalent Relations

Partition

A partition of set A is a set of mutually disjoint **non-empty** subsets of A whose union is A.

Give a partition of set A, the relation induced by the partition, R, is defined as follows: $\forall x, y, \in A, x\ R\ y \Leftrightarrow \exists$ subsets Ai of the partition, $x, y \in Ai$.

Example

Let $A = \{0, 1, 2, 3, 4\}$ and consder the following partition of A: $\{0, 3, 4\}, \{1\}, \{2\}$.
Find the relation R induced by this partition.
Answer: $R = \{(0, 0), (0, 3), (0, 4), (1, 1), (2, 2)\}$

Theorem

Let A be a set with partition and let R be the relation induced by the partition. Then R is *reflexive, symmetric, and transitive.*

Proof: Suppose A is a set with partition $\{A_1, A_2, \ldots, A_n\}$. Without loss of generality, we'll assume a finite partition. Then $\forall i, j \in \{1, 2, \ldots, n\}, i \neq j \Rightarrow A_i \cap A, = \not{0}$. And $\bigcup_{i=1}^{n}$. The relation R induced by the partition is defined as follows: $\forall x, y \in A, x\ R\ y \Leftrightarrow \exists Ai$ of the partition, $x, y \in Ai$.

Reflexive Let $x \in A$, Since $A_1, A_2, \ldots A_n$ is a partition of A, $\exists i \in \{1, 2, \ldots n\}, x \in Ai$. But then the statement $\exists Ai$ of the partition, $x \in Ai$ and $x \in Ai$ is true. Thus, by definition of R, $x\ R\ x$.

Symmetric Let $x, y \in A, x\ R\ y$. Then $\exists Ai$ of the partition, $x \in Ai \wedge y \in Ai$. It follows that the statement $\exists Ai$ of the partition, $y \in Ai$ is true. Hence, by definition of R, $x\ R\ y$.

Transitivity Let $x, y, z \in A, xRy \wedge yRz$. By definition of R, $\exists Ai, A$, of the partition $x, y \in A \wedge y, z \in A$. Suppose $Ai \neq A$, then $Ai \cap A, = \not{0}$. Since $\{A, A_2, \ldots, A_n\}$ is a partition of A. But $y \in Ai \bigwedge y \in Aj$,. Hence $Ai \cap Aj\ \neq \not{0}$. That's a contradiction, thus Ai = A. It follows that $x, y, z \in Ai$. Thus, by definition of R, xRz.

Let A be a set and R a relation on A, R is an <u>equivalence relation</u> iff R is reflexive, symmetric, and transitive.